

Institutional Identity and History

(From Official University of Karachi Department of Computer Science Page — uok.edu.pk/faculties/computerscience and Scribd — Department of Computer Science UBIT Official Document, September 2025)

The Umaer Basha Institute of Information Technology, commonly known as UBIT, is the Department of Computer Science at the University of Karachi. It is one of the most prominent public sector computing institutes in Karachi and Pakistan, operating under the umbrella of the University of Karachi, which itself is a large research university established in 1951, located on University Road, Karachi, spanning a campus of approximately 1,200 acres.

History: The Department of Computer Science was established by a resolution of the Academic Council of the University of Karachi in a meeting held on November 27, 1984, and began functioning in the academic year 1985 to 1986 by offering a degree program in Master of Computer Science, making it one of the first institutions in Karachi to impart education in Computer Science and Technology. The Bachelor of Science in Computer Science morning program was launched in 1996, and the evening program followed in 2001. The first batch of the Bachelor of Science Computer Science morning program passed in 1999. The department was subsequently elevated and renamed the Umaer Basha Institute of Information Technology following a donation by the Umaer Basha Foundation, which jointly sponsored the construction of the dedicated UBIT building on the main campus of the University of Karachi.

The Umaer Basha Foundation and the University of Karachi jointly upgraded the existing Department of Computer Science into the institute now recognized as UBIT. The UBIT complex is comprised of three blocks and houses state-of-the-art computer facilities with leased line internet circuits. The facility includes several lecture rooms, discussion rooms, computer laboratories, and faculty offices, as well as dedicated rooms for teleconferencing, discussions, and training sessions on specialized topics.

Recognition and Rankings: The University of Karachi holds Higher Education Commission rank 2 out of 13 in the General Large category of Pakistani universities. UBIT, as its Department of Computer Science, has published over 300 research articles and is widely regarded as one of the leading public sector institutes for Computer Science and Software Engineering in Karachi. The department also conducts the examination and laboratory sessions for the Bachelor of Computer Science program of affiliated colleges under the University of Karachi.

Campus Location

(From Official University of Karachi Department of Computer Science Page — uok.edu.pk/faculties/computerscience and studentscorner.edu.pk — UBIT Campus Information)

The Umaer Basha Institute of Information Technology is located within the Faculty of Science, University of Karachi, Gulshan-e-Iqbal, Karachi, Sindh, Pakistan. The contact landline is plus 92-21-99261300 to 06, Extension 2462 and 3462, with a fax number of plus 92-21-99261330.

Academic Programs

(From Official University of Karachi Department of Computer Science Programs Page — uok.edu.pk/faculties/computerscience/bs.php and Scribd — Department of Computer Science UBIT Official Document, September 2025)

UBIT currently offers the following programs across undergraduate, postgraduate, and doctoral levels.

Undergraduate Programs

Bachelor of Science in Computer Science is offered in both the Morning and Evening shifts. It is a four-year degree program divided into eight semesters, with two semesters per year. Each semester spans approximately 16 weeks. Students take twelve courses per year, with six courses per semester, each carrying three credit hours.

Bachelor of Science in Software Engineering is offered in the Evening shift.

Bachelor of Science in Artificial Intelligence is also offered by the department. The curriculum covers foundations of mathematics, linear algebra, statistics, computing, automated reasoning, modeling, case studies, machine learning, natural language processing, artificial neural networks, large language models, deep neural networks, graph neural networks, and generative artificial intelligence.

The mission of the Bachelor of Science Software Engineering program is to prepare software engineers for professional careers, advanced study, leadership roles, and entrepreneurial endeavors through exposure to industrial practices, modern teaching, and experiential learning that helps students develop problem-solving skills so they can fully and ethically participate in a diverse society.

Postgraduate Programs

Master of Computer Science is offered with 70 seats. Eligibility requires a Bachelor of Computer Science from institutions affiliated with the University of Karachi, or an equivalent qualification, with a minimum Cumulative Grade Point Average of 2.5.

Master of Science and Doctor of Philosophy programs are also offered by the department in Computer Science, providing an advanced research pathway for students wishing to pursue academic and research careers.

Postgraduate Diploma

A Postgraduate Diploma in Computer and Information Sciences is offered through the evening program, catering to working professionals seeking specialized credentials without committing

to a full master's degree.

Admission Policy (2025-2026)

(From Official University of Karachi Department of Computer Science Page — uok.edu.pk/faculties/computerscience, entrytest.com — UOK BS Computer Science Eligibility 2025-2026, and uokubit.blogspot.com — UBIT Admission Policy)

Eligibility Criteria

For the Bachelor of Science programs at UBIT, the following academic qualifications are accepted. Students must have passed the Higher Secondary Certificate with Mathematics or an equivalent qualification with a minimum of 55% marks. Students holding a Diploma of Associate Engineering in Mechanical, Electrical, or Electronics disciplines are also eligible, provided they have obtained a minimum of 65% marks. Students who are currently registered for Higher Secondary Certificate Part Two or equivalent supplementary examinations and are awaiting their results are also eligible to apply, subject to confirmation of eligibility upon declaration of results.

For the Master of Computer Science program, applicants must hold a Bachelor of Computer Science from institutions affiliated with the University of Karachi, or an equivalent qualification, with a minimum Cumulative Grade Point Average of 2.5.

Admissions to all Bachelor of Science and Master of Computer Science programs are granted on the basis of a mandatory aptitude test conducted by the department. Admission to the Master of Science and Doctor of Philosophy programs follows University of Karachi postgraduate admission regulations, which include evaluation of prior academic performance and the National Testing Service Graduate Assessment Test scores as per Higher Education Commission guidelines.

Merit Formula

The minimum qualifying marks in the entry test for the Bachelor of Science program is 50%. Any candidate who does not achieve 50% in the entry test is not eligible for admission regardless of their academic record.

The merit list for final selection of candidates who qualify in the entry test is prepared based on a combined weightage of the entry test score and the Higher Secondary Certificate or equivalent examination result. According to the University of Karachi's broader admissions formula for departments with entry tests, merit is calculated as 60% weightage on the Higher Secondary Certificate or equivalent result and 40% weightage on the entry test score. The final merit cutoff is determined by the university based on seat availability and applicant performance each cycle.

Admission Categories — KSP Policy

UBIT, as a department of the University of Karachi, follows the university-wide Karachi, Sindh, Pakistan geographical priority system, commonly known as the KSP policy, for distributing open

merit seats.

Category Karachi covers applicants who passed their prerequisite examination from the Karachi Board, as well as those who passed from the Federal Board provided the college where they studied is located within Karachi.

Category Sindh covers applicants who passed their prerequisite examination from any Board located within the Province of Sindh, excluding Karachi.

Category Pakistan covers applicants who passed their prerequisite examination from any Board or teaching institution located in Pakistan or abroad, excluding those located within the Province of Sindh.

Seats are filled in priority order, with Karachi category applicants accommodated first, followed by Sindh category applicants if seats remain, and Pakistan category applicants thereafter.

Admission Test Month and Schedule

The admission test for UBIT is typically held in the last week of November. Admission applications close in the month of November, and test dates are confirmed after application submissions close. The University also accepts National Testing Service results for admission purposes. The University of Karachi conducts its BS admissions entry test at the main campus, and for the 2025 cycle the test was scheduled for July 28, 2025.

Entry Test Details

(From Quora — UBIT Admission Eligibility, entrytest.com — UOK BS Computer Science Test Pattern 2025, and uokubit.blogspot.com — UBIT Entry Test Details)

The UBIT entry test is a 100-mark aptitude examination covering five subject areas. The exact subject composition depends on the educational background of the applicant.

For students who passed the Higher Secondary Certificate under the Pre-Engineering group, the test consists of 20 marks from English, 20 marks from Chemistry, 20 marks from Mathematics, 20 marks from Physics, and 20 marks from General Knowledge.

For students who passed the Higher Secondary Certificate under the Intermediate Certificate of Science group or equivalent with Computer Science as a subject, the Chemistry section is replaced by Computer Science, retaining the same distribution of 20 marks per subject across English, Computer Science, Mathematics, Physics, and General Knowledge.

The minimum passing marks for the entry test is 50 out of 100. Students must appear in the test on campus and are required to bring their admit card, original National Identity Card or Bay Form, a paid fee voucher, basic stationery, and a calculator.

Fee Structure

(From University of Karachi Official Semester Fee Portal — uok.edu.pk/semesterfees, University

As a department of the University of Karachi, UBIT's fee structure falls under the university's broader fee framework, which is approved by the Board of Directors and classifies programs under morning and evening categories.

Morning Program Fees

The estimated fee for the Bachelor of Science Computer Science morning program at the University of Karachi is approximately Rs. 35,000 per semester as of 2025. This figure is drawn from the University of Karachi's Appendix-M fee schedule, which places Computer Science and related programs in the elite tier at Rs. 86,000 per semester, inclusive of an embedded admission fee of Rs. 1,000. Students are advised to verify the current semester fee directly through the university's official semester fee portal at uok.edu.pk/semesterfees, where fee vouchers can be downloaded using their roll number.

Evening Program Fees

The estimated fee for the Bachelor of Science Computer Science evening program is approximately Rs. 55,000 to Rs. 104,500 per semester depending on the applicable fee schedule and batch year. The evening program carries a significantly higher fee than the morning program, consistent with the university-wide evening premium applied across all departments.

The University of Karachi's evening fee structure includes mandatory upfront administrative charges at the time of admission consisting of an admission fee of Rs. 4,500, an enrolment fee of Rs. 1,500, a development fee of Rs. 2,500, and a refundable caution money deposit of Rs. 1,500, bringing the total initial amount to Rs. 10,000 in addition to the base tuition fee.

A one-time registration fee of Rs. 2,200 is payable separately at the time of first enrollment, in addition to the semester fee.

Fee Payment Methods

Fee vouchers can be downloaded from the University of Karachi's semester fee portal using the Quick Pay system powered by UBL. Payment can be made at any UBL branch, through the Quick Pay mobile application, or at the on-campus Alfalah Bank branch. Students must download and present their fee voucher when making cash payments at bank branches.

The university mandates that students of the morning program are not permitted to sit in examinations until all outstanding dues have been cleared. Admit cards are not issued to students with outstanding fee defaults.

Important Note on Fee Verification: Due to the University of Karachi's annual 10% compounding fee escalation policy mandated from the 2025 batch onward, fee figures are subject to increase each academic year. Students are strongly advised to confirm the current fee from the official university semester fee portal before making financial plans.

Refund Policy

(From University of Karachi Official Fee Structure — uok.edu.pk and Higher Education Commission National Fee Refund Policy)

UBIT follows the Higher Education Commission National Fee Refund Policy as applied by the University of Karachi across all departments.

Up to the 7th or 10th day of the start of classes, that is approximately the first week: 100% refund of tuition and semester fees.

Up to the 15th day of the start of classes, that is approximately the second week: 50% refund.

After the 15th day: No refund.

The admission fee component is non-refundable under all circumstances. Students who were admitted conditionally while awaiting their Higher Secondary Certificate results and subsequently do not qualify for final admission receive a 100% refund of the semester fee within three weeks of the announcement of their result.

Scholarships and Financial Assistance

(From Official University of Karachi Scholarships Information and vstudents.org — UOK Scholarship Details 2025)

UBIT students, as members of the University of Karachi, are eligible for all scholarships offered by the parent university. The university allows full and half concessions in the tuition fee to deserving students as per established rules.

Higher Education Commission Need-Based Scholarship: Available to undergraduate students who meet the financial need criteria set by the Higher Education Commission of Pakistan.

Ehsaas Undergraduate Scholarship Program: A federal initiative that can cover up to 100% of tuition fees for financially deserving students from low-income backgrounds.

Internal Departmental Merit Scholarships: Awarded based on strong academic performance within the university's own scholarship framework.

Sindh Educational Endowment Fund Scholarship: Available to students with Sindh domicile on a need-cum-merit basis, covering tuition fees for qualifying students.

Foreign Scholarships: Students of the University of Karachi across various departments have availed foreign scholarships for higher studies, and UBIT students may pursue such opportunities through the relevant university office.

Campus Facilities at UBIT

(From Scribd — Department of Computer Science UBIT Official Document, September 2025 and uokubit.blogspot.com)

Computer Laboratories: The UBIT complex houses multiple state-of-the-art computer laboratories equipped with modern hardware and software resources. The facilities include

leased line internet circuits providing high-speed connectivity for research and academic work.

Research Infrastructure: The institute has dedicated rooms for teleconferencing, specialized training sessions, and research discussions. UBIT has hosted workshops with major international technology organizations including Intel Corporation and a seminar on Cybercrime and Associated Laws in collaboration with the National Response Center for Cyber Crimes of the Federal Investigation Agency.

Faculty: UBIT maintains a team of highly qualified faculty members with advanced degrees and ongoing research engagement. The University of Karachi employs over 893 academics across all departments. UBIT's faculty specializes in machine learning, cybersecurity, data science, software engineering, and artificial intelligence.

Thesis and Research Projects: Final year thesis at the Bachelor of Science level is assigned under course codes BSCS-619 and BSCS-620 exclusively to students who achieve a Cumulative Grade Point Ratio greater than 3.00 up to the sixth semester. This ensures that research projects are handled by academically high-performing students.

Student Life: UBIT encourages student participation in technical competitions, coding events, seminars, and collaborative projects with information technology companies. The department has links with industry organizations providing students with opportunities for internships, professional networking, and industry-relevant project work.

Library: Students have access to the University of Karachi's central library resources, digital library, and the departmental reading facilities within the UBIT building.

Dignitaries and Recognition: UBIT has been visited by prominent government and academic figures including the Governor of Sindh, former Chairman of the Higher Education Commission, and international scholars from South Korea, Italy, Singapore, and senior faculty from leading Pakistani universities including the Lahore University of Management Sciences, COMSATS, the University of Sindh, and Mehran University of Engineering and Technology.

Grading System

(From uokubit.blogspot.com — UBIT Semester Rules and Grading)

The Bachelor of Science four-year degree program at UBIT uses a numerical and alphabetical grading system with a Cumulative Grade Point Ratio formula. A minimum of 50% marks is required to pass any course. Both theory and practical examinations are treated as separate passing heads, meaning a student must pass each component independently.

Summary and Context

(From Official University of Karachi Department of Computer Science Page — uok.edu.pk/faculties/computerscience, thepakistanconnect.com — UBIT Admissions 2026, and techjuice.pk — UBIT University Admissions and Programs)

The Umaer Basha Institute of Information Technology occupies a unique position in Karachi's higher education landscape as the computing faculty of the University of Karachi, one of Pakistan's oldest and largest public universities. As a public sector institution, UBIT offers computing education at a fraction of the cost of private universities while maintaining strong academic standards, a growing research output exceeding 300 published articles, and a faculty base deeply engaged in emerging fields including artificial intelligence, machine learning, cybersecurity, and data science.

The department's dual morning and evening program structure makes it accessible to both full-time students and working professionals. Its placement under the University of Karachi also means students benefit from the parent university's extensive affiliated college network, scholarship programs, and institutional credibility. The KSP geographical admission policy, while currently under legislative scrutiny for reform across the university, continues to govern seat distribution at UBIT in the 2025 admission cycle. The mandatory entry test with a minimum 50% passing threshold ensures a baseline standard of academic aptitude is applied universally to all incoming Bachelor of Science students regardless of their previous examination results.

Graduates of UBIT pursue careers as software developers, engineers, information technology consultants, database administrators, network engineers, research scientists in artificial intelligence and machine learning, and academic roles in universities and colleges. A significant number of UBIT alumni have secured positions in leading technology companies, government organizations, and research institutions both in Pakistan and internationally.